Teaching with AI-Generated Scenarios

Prompt: Act as a Cisco CCNA Instructor. Create five routing and switching scenario-based configuration output questions that students will have to examine, make a hypothesis, and determine the correct answer and how to fix it. Think deeply about this.
Before you respond, create an internal rubric for what defines a 'world-class' answer to my request. Then internally iterate on your work until it scores 10/10 against that rubric, and show me only the final, perfect output.
Example ideas:
'''
In-Class Use Ideas:

- Use scenarios as review warm-ups or quiz questions
  - *"What's the correct answer—and how do you know?"*
- Remove the explanation and have students justify their answer
  - Builds critical thinking and CCNA exam prep skills
- Have students write their own prompt to generate practice questions
  - Swap with peers to solve and discuss

Teaching Tips:

- Focus on reasoning, not just recall
- Let students compare AI's logic to their own
- Use AI to remix scenarios by changing addresses, ports, or protocols

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Researching 48 sources read

- Research current CCNA routing and switching exam topics and objectives
- Investigate common network configuration issues and troubleshooting methodologies in Cisco environments
- Find examples of scenario-based questions used in CCNA certification preparation
- Research best practices for creating educational assessment questions that test critical thinking
- Look for rubrics and quality criteria for technical education materials in networking

Response
Five World-Class CCNA Routing and Switching Scenario-Based Configuration Output Questions

Internal Rubric for a 'World-Class' Answer
1. Alignment with CCNA Objectives:
Scenarios must directly reflect current CCNA 200-301 exam topics, including routing, switching, VLANs, OSPF, ACLs, and troubleshooting methodologies .
2. Realism and Relevance:
Each scenario should mirror real-world network issues, using authentic Cisco CLI outputs and plausible misconfigurations .
3. Critical Thinking Focus:
Questions must require students to analyze, hypothesize, and justify their answers, not just recall facts. Each should prompt reasoning about root causes and solutions .
4. Clarity and Completeness:
Scenarios must be clearly worded, with all necessary information provided, and free from ambiguity .
5. Depth and Variety:
Scenarios should cover a range of topics (routing, switching, VLANs, ACLs, etc.) and cognitive levels (analysis, synthesis, evaluation) .

6. Actionable Solutions:
Each scenario should have a clear, correct answer and a specific, actionable fix that students can articulate .
7. Assessment Value:
Questions should be suitable for in-class discussion, quizzes, or exam prep, and encourage students to explain their reasoning .
8. Technical Accuracy:
All configurations and outputs must be technically correct and up-to-date with current Cisco best practices .
9. Engagement:
Scenarios should be interesting and challenging, promoting engagement and deeper learning .
10. Iterative Perfection:
Each scenario is reviewed and refined for maximum instructional value, clarity, and challenge.

Final Output: Five Scenario-Based Configuration Output Questions

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Scenario 1: Inter-VLAN Routing Issue
Background:
A network uses Router-on-a-Stick for inter-VLAN routing. Users in VLAN 10 cannot reach users in VLAN 20. Below is the relevant switch and router configuration output:
Switch Output:

```
Switch# show vlan brief
VLAN Name                             Status    Ports
---- -------------------------------- --------- --------------
1    default                          active    Fa0/1, Fa0/2
10   Sales                            active    Fa0/3, Fa0/4
20   Engineering                      active    Fa0/5, Fa0/6

Switch# show interfaces trunk
Port      Mode         Encapsulation  Status        Native vlan
Fa0/24    on           802.1q         trunking      1
```

Router Output:

```
Router# show running-config | section interface
interface FastEthernet0/0
 no ip address
!
interface FastEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
!
interface FastEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
```

Question:
Examine the outputs. What is the most likely cause of the inter-VLAN routing failure? How would you fix it?

Scenario 2: OSPF Routing Table Discrepancy
Background:
Three routers (R1, R2, R3) are configured for OSPF in area 0. R1 and R2 see all routes, but R3 is missing the route to 10.1.2.0/24. Below is the relevant output from R3:

```
R3# show ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
192.168.12.1    1     FULL/DR         00:00:33    192.168.12.1    Gig0/0

R3# show running-config | section ospf
router ospf 1
 network 192.168.12.0 0.0.0.255 area 0
```

Question:
Based on the output, hypothesize why R3 is missing the route to 10.1.2.0/24. What configuration change would resolve the issue?

Scenario 3: Switchport VLAN Assignment Error
Background:
A user connected to port Fa0/7 on Switch1 cannot access the network. The network uses VLANs for segmentation. Below is the relevant output:

```
Switch1# show interfaces status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/7                        connected    1          a-full  a-100
10/100BaseTX

Switch1# show vlan brief
VLAN Name                             Status    Ports
---- -------------------------------- --------- ----------------------------
---
1    default                          active    Fa0/1, Fa0/2, Fa0/7
10   Sales                            active    Fa0/3, Fa0/4
20   Engineering                      active    Fa0/5, Fa0/6
```

Question:
Analyze the outputs. What is the likely cause of the user's connectivity issue? What steps would you take to fix it?

Scenario 4: ACL Blocking Legitimate Traffic
Background:
An access control list (ACL) is applied to a router interface to restrict access. Users report they cannot access the web server at 192.168.30.10. Below is the ACL and interface configuration:

```
Router# show running-config | include access-list|interface
interface GigabitEthernet0/1
 ip address 192.168.20.1 255.255.255.0
 ip access-group 101 in

access-list 101 deny tcp any host 192.168.30.10 eq 80
access-list 101 permit ip any any
```

Question:
Evaluate the configuration. Why are users unable to access the web server? How would you modify the configuration to allow legitimate access?

Scenario 5: Spanning Tree Port Blocking
Background:
A network with redundant links between two switches is experiencing connectivity issues. Below is the spanning tree output from SwitchA:

```
SwitchA# show spanning-tree vlan 1
VLAN0001
  Root ID    Priority    32769
             Address     0011.2233.4455
             Cost        19
             Port        1 (FastEthernet0/1)
  Bridge ID  Priority    32769
             Address     0011.2233.5566
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- -------------------------------
-
Fa0/1            Root FWD 19        128.1    P2p
Fa0/2            Altn BLK 19        128.2    P2p
```

Question:
Interpret the output. Why might some devices be experiencing connectivity issues? What change could you make to optimize redundancy and connectivity?

Teaching and Assessment Tips

- **Use as Warm-Ups or Quizzes:** Present scenarios and ask students to hypothesize, justify, and propose solutions.
- **Remove Explanations for Peer Discussion:** Have students explain their reasoning and compare answers.
- **Encourage Student-Generated Scenarios:** Deepens understanding and critical thinking.
- **Focus on Reasoning:** Ask students not just for the fix, but for the logic behind their diagnosis.

These scenarios are designed to challenge students’ analytical skills, reflect real-world CCNA exam and job requirements, and foster deep understanding of routing and switching concepts.